

Introduction to L^AT_EX

Write documents like a scientist

Guest lecture for B.Tech in Artificial Intelligence
Global College of Social Science & Technology, Kathmandu
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Slides, notes, exercises
github.com/danish-puri/LaTeX

Try it in five minutes

1. Go to overleaf.com and register. The free account is all you need.
2. Click **New Project**, then **Blank Project**.
3. Replace the body with the skeleton below.
4. Press **Recompile** (Ctrl+Enter, or Cmd+Enter on a Mac).
5. The PDF appears on the right. Edit, recompile, repeat. That loop is the whole workflow.

```
\documentclass{article}
\begin{document}
Hello, \LaTeX! Math sits right
in a sentence,  $e^{i\pi}+1=0$ .
\end{document}
```

Watch out. The characters % & _ # \$ have special jobs. To print one, escape it, so \% \& _ \# \\$. A bare % starts a comment. L^AT_EX ignores the rest of that line, which is also how you leave notes to yourself.

Math mode essentials

Wrap math in $...$$ inside a sentence, or in $\[...\]$ to give it its own line.

Type this	Get this
x^2	x^2
a_{ij}	a_{ij}
$\frac{a}{b}$	$\frac{a}{b}$
$\sqrt{x+y}$	$\sqrt{x+y}$
α, θ, π	α, θ, π
$\sum_{i=1}^n i$	$\sum_{i=1}^n i$
$\int_0^1 x \, dx$	$\int_0^1 x \, dx$
$\lim_{x \rightarrow 0} f(x)$	$\lim_{x \rightarrow 0} f(x)$

The $\,$, in the integral row is a thin space.

You already know enough to typeset real machine learning. Gradient descent is one line.

$$\theta_{t+1} = \theta_t - \eta \nabla L(\theta_t)$$

So is softmax.

$$\text{softmax}(z_i) = \frac{e^{z_i}}{\sum_j e^{z_j}}$$

Everyday commands

Command	What it does
$\usepackage{...}$	loads an add-on package
$\section{...}$	numbered heading
$\textbf{...}$	bold
$\textit{...}$	<i>italics</i>
$\itemize$ env	bullet list
$\enumerate$ env	numbered list
$\includegraphics$	insert an image (needs graphicx)
$\label + \ref$	numbers that fix themselves
$\tableofcontents$	builds itself
$\cite{...}$	citations via BibTeX

When it breaks

- It will break. That is normal, not a sign you are bad at this.
- Read the **first** error in the log, not the last. Errors cascade. The line number points at or just after the real problem.
- Compile often. Then the bug is always in the last thing you typed.

The three errors you will meet first, translated.

The log says	It means
Undefined control sequence	misspelled command
Missing \$ inserted	math outside $...$$
$\begin{...}$ ended by $\end{...}$	mismatched environment

Your challenge

Pick one page of any course handout, lab report, or your own notes, and rebuild it in L^AT_EX. The tables on this page cover everything you need. Stuck on something? Open an issue on the repo and I will answer.

Keep learning

- Scan the QR code above. The repo has my slides, lecture notes, homework, and a cheat sheet.
- overleaf.com/learn has the best free tutorials.
- Need a full paper, thesis, or resume? Never start from scratch. Steal a template from [the Overleaf gallery](#).
- AI assistants write decent L^AT_EX. Let them type, but read what they produce and fix it. You are the editor.